

Ammar Nagri

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EDUCATION

The University of Manchester

Manchester, GB

Bachelor of Science in Computer Science and Mathematics

Sep 2025 - June 2028

- Projected First Class Honours: Probability I, MFA, Statistics I, Data Science, Intro to Programming I (Python), Intro to Programming II (Java), First Year Team Project

EXPERIENCE

Software Engineer

July 2026 – Sept-2026

Mubit-AI

London, GB

- Optimised memory retrieval by replacing an $O(\text{session size})$ scan with a near $O(\text{limit})$ reverse scan, achieving up to $2,500\times$ faster limited-history lookups while preserving identical behaviour and chronological ordering.
- Improved storage by implementing tiered per-column-family Zstd compression (reducing disk usage by 35% with minimal latency impact)

Machine learning Intern

Jan 2026 – April 2026

Level2

New York, NY (Remote)

- Built a **gaussian HMM classifier** for intra-day market data's regime
- Developed a **live regime forecaster** using **temporal convolutional networks** and **PatchTST** architectures, utilising **Monte Carlo dropout** for Bayesian uncertainty decomposition, achieving an **AUC of 76.1%** on SPY
- Presented my findings with fellow students and am currently working on presenting it to more financial startups

Software Engineer

Aug 2022 – Aug 2022

JustEat Takeaway (formerly Flyt)

London, GB

- Conducted Go benchmarking and pprof CPU/memory profiling to **optimise** a core microservice in JustEat, achieving $\sim 4\times$ **latency improvement**, $\sim 3\times$ **fewer allocations**, and $\sim 2\times$ **reduction in RSS memory usage**
- Designed and implemented a HTTP endpoint for an order analytics microservice, handling query parsing, **SQL DB access**, and robust error handling, defined via OpenAPI specifications and supporting $\sim 700,000$ daily requests
- Developed unit tests using httptest and validated functionality through Postman-based integration testing

Project leader

Sep 2025 – Present

The University of Manchester - Manchester AI Society

Manchester, GB

- Led workshops to teach students machine learning, averaging **75+ students** with a **76% retention rate**
- Architected and built an authoritative real-time multiplayer backend for poker using **FastAPI** and **WebSockets**
- Optimised core components, achieving up to **4M+ ops/s** in rule evaluation and allowing **40+** concurrent clients

PROJECTS

Changes in Stock Market Structure | *Python, pandas, NumPy, scikit-learn, matplotlib, seaborn*

June 2026

- Built a data pipeline to ingest S&P 500 historical constituents and Polygon market data, filtering prices to each stock's membership window before computing daily return correlation matrices and sector-level heatmaps
- Modelled evolving market structure using **Graphical Lasso** sparse partial-correlation networks and **Affinity Propagation** clustering, visualising annual changes with **MDS-based** network layouts and animated frames

Particle Collision Data Analysis with ATLAS | *NumPy, pandas, Jupyter, Git, matplotlib*

Sept 2023 – June 2024

- Led a team of 5 students to reconstruct the Z^0 **boson** and calculate its invariant mass using NumPy and Pandas
- Attended a masterclass at the **University of Oxford**, presenting and defending our methodology and findings
- Co-authored a research poster and delivered a poster presentation at a London physics conference

COMPETITIONS AND AWARDS

International Olympiad of AI (Apr 2025): Silver based on examination performance

UKMT Junior, Intermediate and Senior Mathematics Challenges (2018–2024): Multiple Gold Awards

TECHNICAL SKILLS

Languages: Python, Java, PostgreSQL, MySQL, Golang, R, TypeScript, C++, JavaScript, LaTeX

Libraries: NumPy, Pandas, matplotlib, scikit-learn, OpenCV, Hugging Face Transformers

Frameworks & Tooling: PyTorch, TensorFlow, FastAPI, Pydantic, Docker, Git, Weights & Biases, AWS, pytest, pprof